Derivation of the Gain Measurement a_{xm} and its Noise Statistics ($\delta x_m \to a_{xm}$)

Measurement

(d-step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k], \qquad d = 2,$$

where $\delta x[k] = x_d[k] - x[k]$ is the tracking error, $n_x[k] \sim \mathcal{N}(0, \sigma_n^2)$ is the sensor noise, and λ_c ($0 \le \lambda_c < 1$) is the closed-loop pole.

Closed-loop error

Control objective: $\delta x[k+1] = \lambda_c\, \delta x[k]$.

Control law (4-state; the gain rate is fed forward, the disturbance is not compensated):

$$f_{dx}[k] = \hat{a}_x^{-1}[k]\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x_m[k] - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i]\, f_{dx}[k-i]\Big]\Big\},$$

with $\Delta x_d^d[k] = \Delta x_d[k] + (1-\lambda_c)\sum_{i=1}^{d} \Delta x_d[k-i]$ the (delay-compensated) desired-trajectory feedforward. Under this law, and dropping the gain-estimation and disturbance terms (which are higher order), the closed-loop error becomes

$$\delta x[k+1] = \lambda_c\, \delta x[k] - \delta x_T^d[k] - (1-\lambda_c)\, n_x[k],$$

$$\delta x_T^d[k] = \delta x_T[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} \delta x_T[k-i], \qquad \delta x_T[k] := a_x[k]\, f_T[k] \;\; (\text{white, } \sigma_{\delta x_T}^2 = 4k_B T\, a_x).$$

Solving this recursion in the z-domain gives two noise channels:

$$\delta x(z^{-1}) = -\, H_T(z^{-1})\, \delta x_T(z^{-1}) \; - \; H_n(z^{-1})\, n_x(z^{-1}),$$

$$H_T(z^{-1}) = z^{-1}\frac{1 + (1-\lambda_c)z^{-1} + (1-\lambda_c)z^{-2}}{1-\lambda_c z^{-1}}, \qquad H_n(z^{-1}) = z^{-1}\frac{1-\lambda_c}{1-\lambda_c z^{-1}}.$$

IIR mean removal

Low-pass (EWMA) mean and high-pass residual:

$$\overline{\delta x}_m[k] = (1-a_{pd})\, \overline{\delta x}_m[k-1] + a_{pd}\, \delta x_m[k], \qquad H_{LP}(z^{-1}) = \frac{a_{pd}}{1-(1-a_{pd})z^{-1}},$$

$$\delta x_r[k] = \delta x_m[k] - \overline{\delta x}_m[k], \qquad H_{HP}(z^{-1}) = 1 - H_{LP}(z^{-1}) = \frac{(1-a_{pd})\big(1-z^{-1}\big)}{1-(1-a_{pd})z^{-1}}.$$

Since $H_{HP}(z{=}1) = 0$, the residual δx_r is exactly zero-mean.

Channel decomposition (substitute $\delta x_m = \delta x[k-d] + n_x$, independent $\delta x_T \perp n_x$):

$$\boxed{\delta x_r(z^{-1}) = -\, F_T(z^{-1})\, \delta x_T(z^{-1}) + F_N(z^{-1})\, n_x(z^{-1})}$$

$$F_T = H_{HP}\, z^{-d}\, H_T, \qquad F_N = H_{HP}\big[\, 1 - z^{-d} H_n \,\big] = (1 - z^{-1})\, z^3\, F_T.$$

Residual variance (by Parseval, residues at the poles $z = 1 - a_{pd}$ and $z = \lambda_c$; $D := 1 - (1 - a_{pd})\lambda_c$):

$$\boxed{\sigma^2_{\delta x_r} = C_{\text{dpmr}}(\lambda_c, a_{pd})\, \sigma^2_{\delta x_T} + C_n(\lambda_c, a_{pd})\, \sigma^2_n}$$

$$C_{\text{dpmr}} = (1 - a_{pd})^2 \left[\frac{2(1 - a_{pd})(1 - \lambda_c)}{D} + \frac{2}{2 - a_{pd}} \cdot \frac{1}{(1 + \lambda_c)\, D} \right],$$

$$C_n = \frac{2(1 - a_{pd})^2}{2 - a_{pd}} \left[1 + \frac{(1 - a_{pd})^2 a_{pd} (1 - \lambda_c)}{D} + \frac{(1 - \lambda_c)^2}{(1 + \lambda_c)\, D} \right] \qquad \text{(full } a_{pd} \text{ closed form).}$$

Variance estimate and gain measurement

EWMA variance of the residual:

$$\hat\sigma^2_{\delta x_r}[k] = (1 - a_{cov})\, \hat\sigma^2_{\delta x_r}[k-1] + a_{cov}\, \delta x_r^2[k], \qquad H_E(z^{-1}) = \frac{a_{cov}}{1 - (1 - a_{cov}) z^{-1}}.$$

Because $H_{HP}(z{=}1) = 0$ (zero-mean residual) and $H_E(z{=}1) = 1$, the estimate is unbiased:

$$\mathbb{E}[\hat\sigma^2_{\delta x_r}] = \sigma^2_{\delta x_r} = C_{\text{dpmr}}\, 4 k_B T\, a_x + C_n\, \sigma^2_n.$$

Linear inversion (linear in $\hat\sigma^2_{\delta x_r}$, not a square root — the variance is itself *linear* in a_x):

$$\boxed{a_{xm}[k] = \frac{\hat\sigma^2_{\delta x_r}[k] - C_n\, \sigma^2_n}{C_{\text{dpmr}} \cdot 4 k_B T}, \qquad \mathbb{E}[a_{xm}] = a_x.}$$

Measurement noise $R(2,2) = \mathrm{Var}(a_{xm})$

Variance scaling. a_{xm} is an affine map of $\hat\sigma^2_{\delta x_r}$ (slope $m = C_{\text{dpmr}} 4 k_B T$, offset $C_n \sigma^2_n$), so the variance scales by $1/m^2$ (this also converts the units, $\mu\text{m}^4 \to (\mu\text{m/pN})^2$):

$$R(2,2) = \mathrm{Var}(a_{xm}) = \frac{\mathrm{Var}(\hat\sigma^2_{\delta x_r})}{(C_{\text{dpmr}} \cdot 4 k_B T)^2}.$$

Variance of the EWMA estimate (three independent layers):

$$\mathrm{Var}(\hat\sigma^2_{\delta x_r}) = K_{\text{var}}\, \sigma^4_{\delta x_r}\, \mathrm{IF}, \qquad K_{\text{var}} = \frac{2 a_{cov}}{2 - a_{cov}},$$

where K_{var} combines Isserlis' fourth moment ($\mathrm{Var}(X^2) = 2\sigma^4$ for Gaussian X) with the EWMA variance gain $a_{cov}/(2 - a_{cov})$, and IF is the color-inflation factor: δx_r inherits the poles $1 - a_{pd}$ and λ_c, so it is autocorrelated and so is δx_r^2,

$$\mathrm{IF} = 1 + 2 \sum_{\tau \ge 1} \rho^2_{\delta x_r}(\tau)\, s^\tau, \quad s = 1 - a_{cov}.$$

The square ρ^2 comes from Isserlis at lag τ: $\mathrm{Cov}(\delta x_r^2[k], \delta x_r^2[k{+}\tau]) = 2\, \mathrm{Cov}(\delta x_r[k], \delta x_r[k{+}\tau])^2 = 2 \rho^2_{\delta x_r}(\tau)\, \sigma^4_{\delta x_r}$. The autocorrelation decays as $\rho_{\delta x_r}(\tau)\big|_{\tau \ge 3} = c_\alpha (1 - a_{pd})^\tau + c_\beta\, \lambda_c^\tau$.

Result. Substituting $\sigma^2_{\delta x_r} = C_{\text{dpmr}} 4k_B T\, a_x + C_n \sigma_n^2$ gives $\sigma^4_{\delta x_r}/(C_{\text{dpmr}} 4k_B T)^2 = (a_x + \xi)^2$, hence

$$\boxed{R(2,2) = K_{\text{var}}\ \text{IF}\ (a_x + \xi)^2, \qquad \xi = \frac{C_n}{C_{\text{dpmr}}} \cdot \frac{\sigma_n^2}{4k_B T}.}$$

The relative noise $\sigma(a_{xm})/a_x \approx \sqrt{K_{\text{var}}\,\text{IF}}$ (thermal-dominated, $\xi \ll a_x$) is an irreducible finite-sample (chi-squared) floor, set by the EWMA window $N_{\text{eff}} \approx 1/a_{cov}$.

Delay term and KF measurement

Gain model (4-state; the gain rate δa_x is fed forward from the desired trajectory, δa_{ram} is the random part):

$$a_x[k+1] = a_x[k] + \delta a_x[k] + \delta a_{ram}[k],$$

$$\delta a_x[k] = a_x'[k]\,\Delta h_d[k] \text{ (known, fed forward)}, \qquad \delta a_{ram}[k] = a_x'[k]\,\Delta x_{ram}[k].$$

The gain measurement reflects the gain d steps earlier; expanding $a_x[k-d]$ and separating the known fed-forward part,

$$y_2[k] = a_{xm}[k] = a_x[k-d] + n_a[k] = a_x[k] - \textstyle\sum_{i=1}^d \delta a_x[k-i] + \underbrace{\left\{ n_a[k] - \textstyle\sum_{i=1}^d \delta a_{ram}[k-i] \right\}}_{r_2[k]}.$$

Since $\sum_{i=1}^d \delta a_x[k-i]$ is known, it is moved to a known correction $\mathbf{c}[k]$, and the state keeps only a_x:

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{c}[k] = \begin{bmatrix} 0 \\ \sum_{i=1}^d \delta a_x[k-i] \end{bmatrix}, \qquad \mathbf{y}[k] = \mathbf{H}\,\mathbf{x}[k] - \mathbf{c}[k] + \mathbf{r}[k].$$

Effective measurement variance (the intrinsic IIR noise n_a and the past random gain increments are independent):

$$R(2,2)_{\text{eff}} = \underbrace{R(2,2)}_{=\text{Var}(n_a)} + \sum_{i=1}^{d} \text{Var}\big(\delta a_{ram}[k-i]\big), \qquad \text{Var}(\delta a_{ram}) = \frac{2}{1+\lambda_c}\left(\frac{a_x K_h}{R}\right)^2 \sigma^2_{\delta h},$$

with $K_h = (1/c)\,(dc/d\bar{h})$ the wall sensitivity, R the probe radius, and $\sigma^2_{\delta h} = 4k_B T\, a_\perp$ the wall-normal thermal step variance. The random-gain term $\delta x^d_{\delta af}$ (the contribution of δa_{ram} to δx) is second order and is omitted from the C_{dpmr} derivation above.